Universidad Nacional de Loja

Faculty of Energy, Industry,

and Non-Renewable Natural Resources

Electrical Engineering Program

Preliminary Design of a Transmission Line from Yanacocha Substation to San Cayetano Substation

Transmission Lines and Electrical Substations

AUTHOR:

Leonardo Gabriel Granda Erazo

INSTRUCTOR:

Eng. Jorge Isaac Peralta Córdova

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Contents

1 Assignment Requirements

The objective is to develop a preliminary design for a transmission line connecting the Yanacocha Substation, operated by Transelectric, to the San Cayetano Substation, operated by Empresa Eléctrica Regional del Sur S.A. (EERSSA), using PLS-CADD. The required activities are as follows:

- Obtain a preliminary route in Google Earth.
- Create the line alignment in PLS-CADD.
- Generate plan, profile, and three-dimensional (3-D) views in PLS-CADD.
- Triangulate the terrain to interpolate points in areas where the elevation value is $Z = 0$.
- Insert satellite imagery over a 400-m-wide alignment corridor.
- Define restricted areas in accordance with the applicable regulations.
- Create a line-support structure based on an actual design rated at 345 kV or above.
- Install a conductor selected from the software library along the entire route. The stringing conditions shall comply with the tension percentages listed in Table 1.
- Verify that the proposed line sections satisfy the operating conditions by reviewing the line report and conductor-utilization results.

Table 1: Design weather cases and conductor tension limits

No.	Weather case	Cable condition	Ultimate tension (%)
1	NESC Medium District Loading (250B)	Initial RS	40
2	NESC Extreme Wind (250C)	Initial RS	60
3	NESC Concurrent Ice and Wind (250D)	Initial RS	60
4	NESC Tension Limit (261H1b)	Initial RS	35
5	NESC Tension Limit (261H1b)	Creep RS	30

2 Development

2.1 Earthdata Search

The design process begins with the acquisition of terrain data from the NASA Earthdata Search platform [1]. The selected dataset is the *ASTER Global Digital Elevation Model V003*. The Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) dataset provides a global digital elevation model (DEM) for terrestrial areas at a spatial resolution of one arc-second, equivalent to approximately 30 m of horizontal spacing at the equator.

After selecting the elevation model, the area covering both candidate substations is defined. Figure 1 shows a Google Earth satellite image used as the geographical reference for the design [2].

Figure 1: Satellite view of the area between the Yanacocha and San Cayetano substations.

The Earthdata Search interface is then used to delimit an appropriate coverage area. The rectangular selection tool is particularly useful for this purpose. Figure 2 shows the area selected for data acquisition.

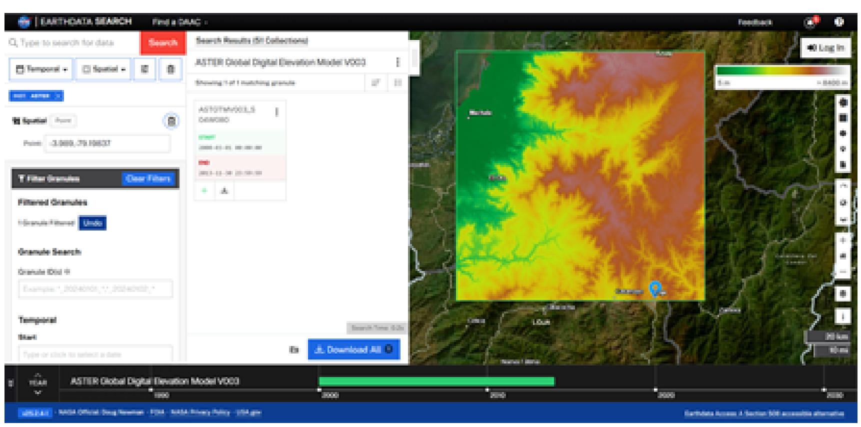


Figure 2: ASTER GDEM dataset and selected download area in Earthdata Search.

Among the available download products, the file with the `dem.tif` extension is selected. Because these data cannot be used directly in PLS-CADD without first defining the coordinate reference system for the study area, the dataset is subsequently processed in Global Mapper.

2.2 Data Parameterization in Global Mapper

The `dem.tif` file is compatible with Global Mapper and is loaded directly into the software, as shown in Fig. 3.

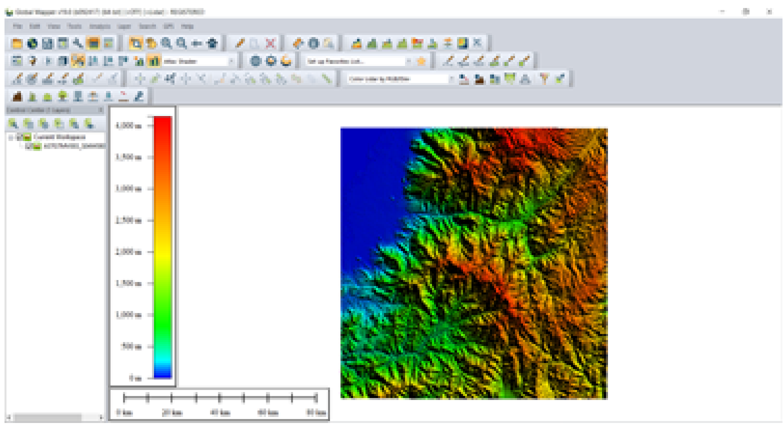


Figure 3: Digital elevation model loaded in Global Mapper.

Ecuador spans four Universal Transverse Mercator (UTM) coordinate zones. The study area is located in UTM Zone 17 South. This projection must be specified to georeference the downloaded points correctly. In Global Mapper, the coordinate system is configured through *Tools* > *Configure* > *Projection*; the adopted parameters are shown in Fig. 4.

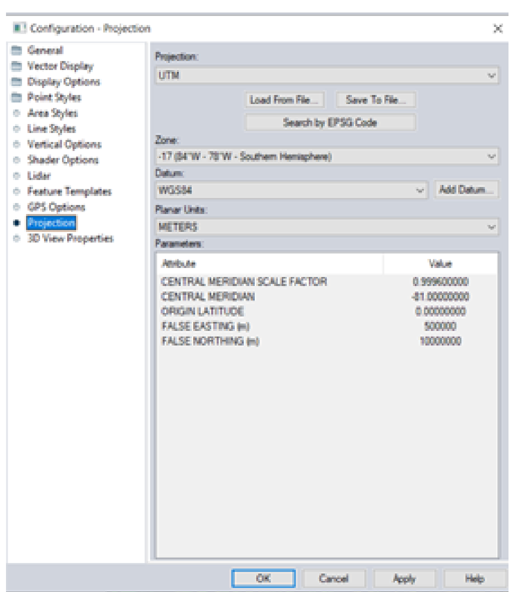


Figure 4: Projection configuration for UTM Zone 17 South in Global Mapper.

Once the coordinate system has been defined, the terrain data are exported in the *PLS-CADD XYZ File* format. The export sequence is shown in Fig. 5.

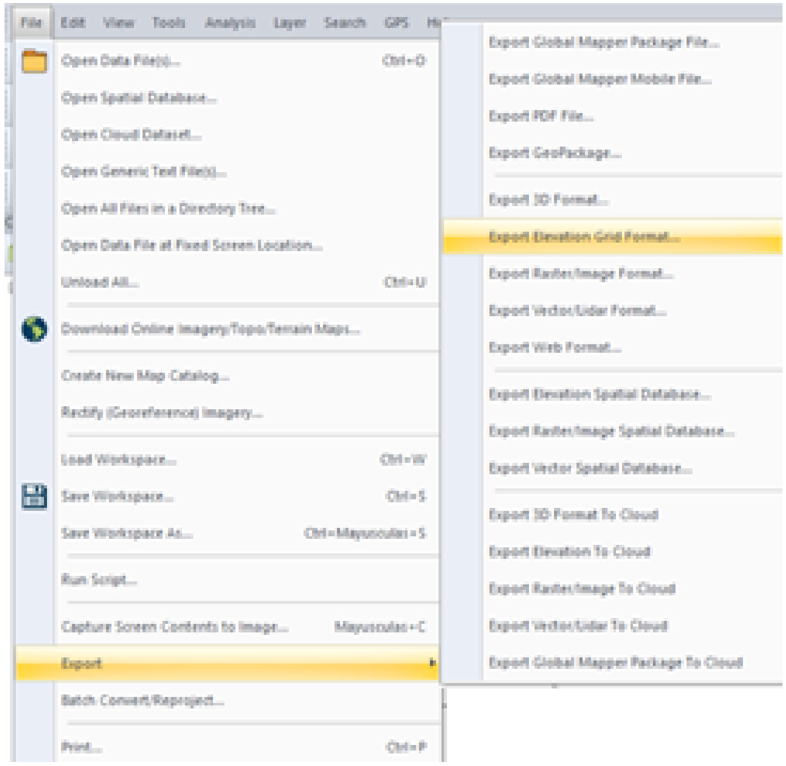


Figure 5: Export of the processed terrain dataset to PLS-CADD XYZ format.

2.3 Preliminary Transmission-Line Route

A preliminary route must be established before the transmission line can be modeled. A new Google Earth project is created for this purpose, as shown in Fig. 6 [2].

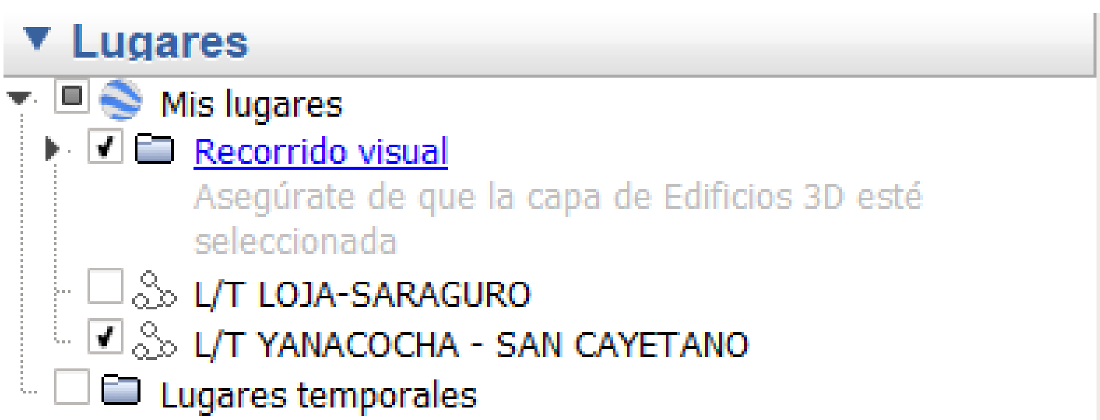


Figure 6: Creation of the preliminary line-route project in Google Earth.

The route is drawn while accounting for restricted areas through which a high-voltage line should not pass. Under the applicable national requirements, transmission-line routing should avoid populated areas, protected areas, roads, and other constrained locations. In this particular study, some constraints cannot be avoided without introducing substantially longer and impractical route segments.

The coordinates of both substations are located and connected by a preliminary alignment. Figure 7 presents the proposed route and its elevation profile.

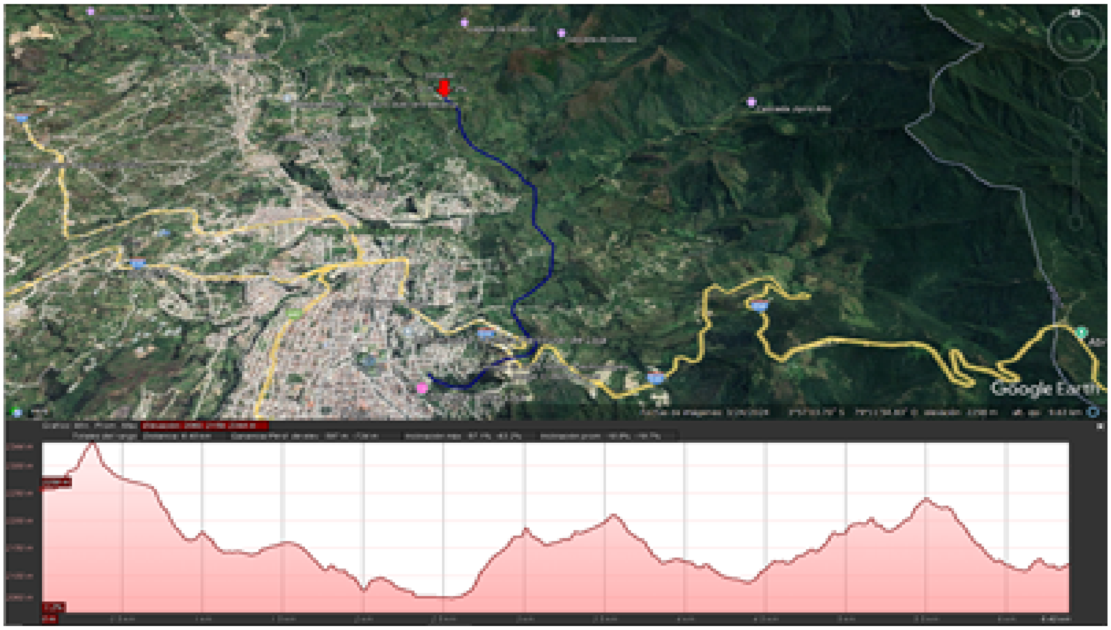


Figure 7: Preliminary transmission-line route and elevation profile in Google Earth.

The route is saved as a `.kmz` file, which is compatible with PLS-CADD and can therefore be used in the detailed line model.

2.4 PLS-CADD Environment

Before the line-route and topographic files are imported, the project coordinate system must be configured in the same manner as in Global Mapper. In the *Terrain* menu, the coordinate-system command is selected to define a project-specific system, as illustrated in Fig. 8.

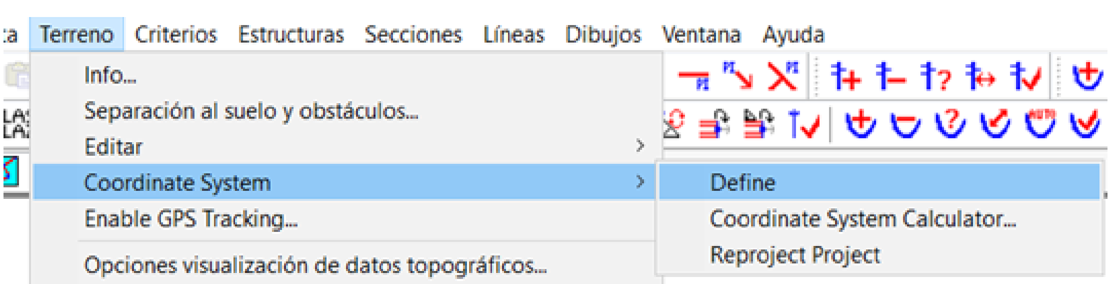


Figure 8: Access to the project coordinate-system definition in PLS-CADD.

The PLS-CADD coordinate parameters are then set to UTM Zone 17 South, using the same settings adopted in Global Mapper (Fig. 9).

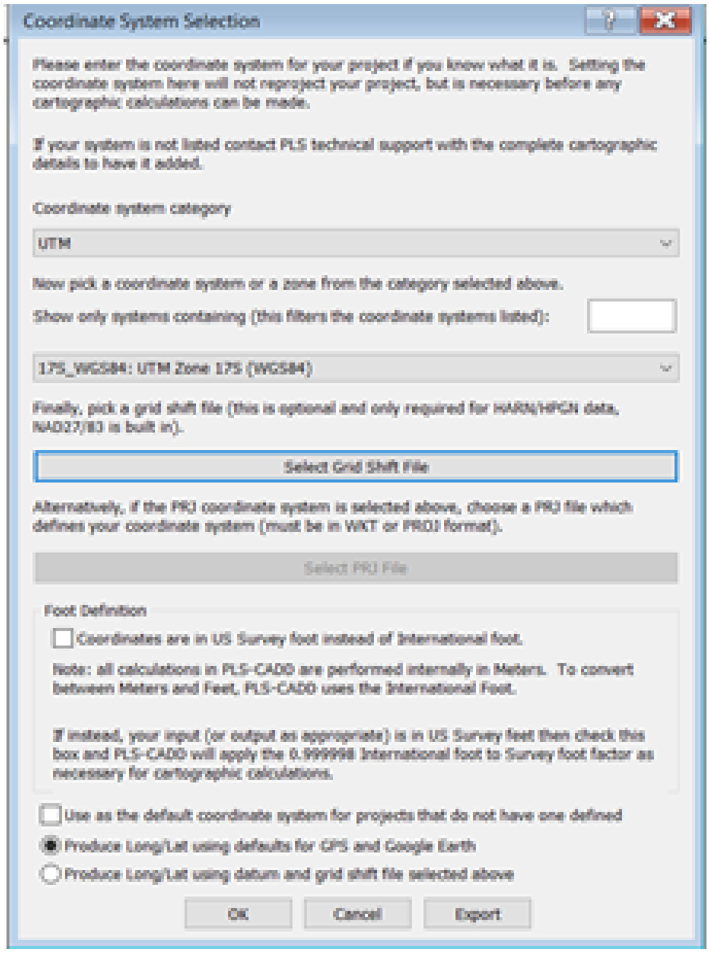


Figure 9: Project coordinate-system parameters in PLS-CADD.

The terrain file is imported using the sequence shown in Fig. 10. In the import dialog, *Reproject Coordinates* is enabled and the project coordinate system is specified again.

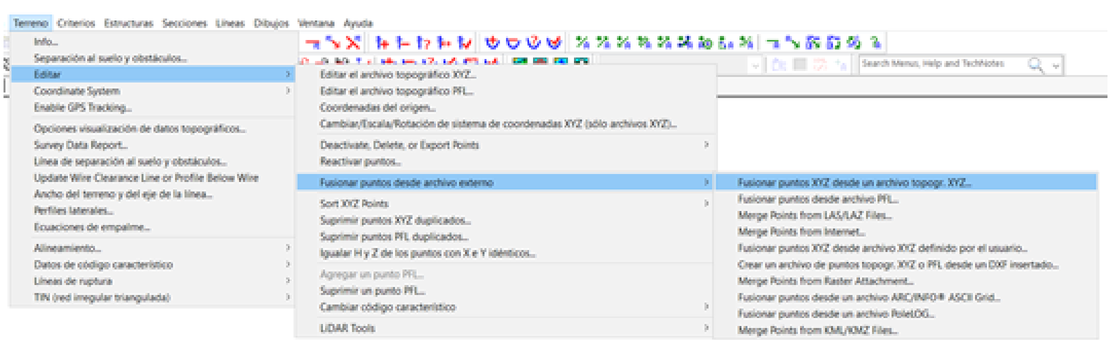


Figure 10: Import and coordinate-reprojection sequence for terrain data.

To display the imported information, the feature-code data are edited from the *Terrain* tool. A 345-kV clearance category is added, and a terrain layer named `Capa_Terreno` is created. In the topographic-data display options, *Draw Only Designed Feature Codes* is selected and the newly created layer is enabled. Pressing *Init* produces the view shown in Fig. 11.

Figure 11: Initial display of the terrain layer imported from Global Mapper.

Figure 12 shows the complete Global Mapper point dataset after initialization.

Figure 12: Imported topographic point cloud in the PLS-CADD workspace.

A second layer, `Capa_PI`, is created for the Google Earth route. In *Terrain* > *Edit* > *Merge*, the KMZ data option is selected and the preliminary route is assigned to this new layer. Both project layers are then enabled in the display options. Figure 14 shows a magnified view of the combined datasets.

The Earthdata dataset contains many points outside the area required for the line model. These points are removed using *Terrain > Edit > Deactivate Inside Polygon*, with the configuration shown in Fig. 13.

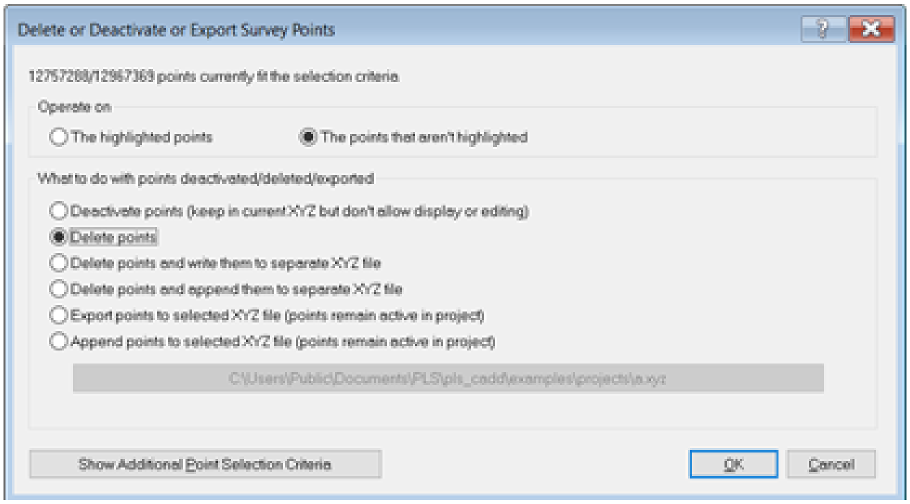


Figure 13: Configuration used to deactivate unnecessary terrain points inside a polygon.

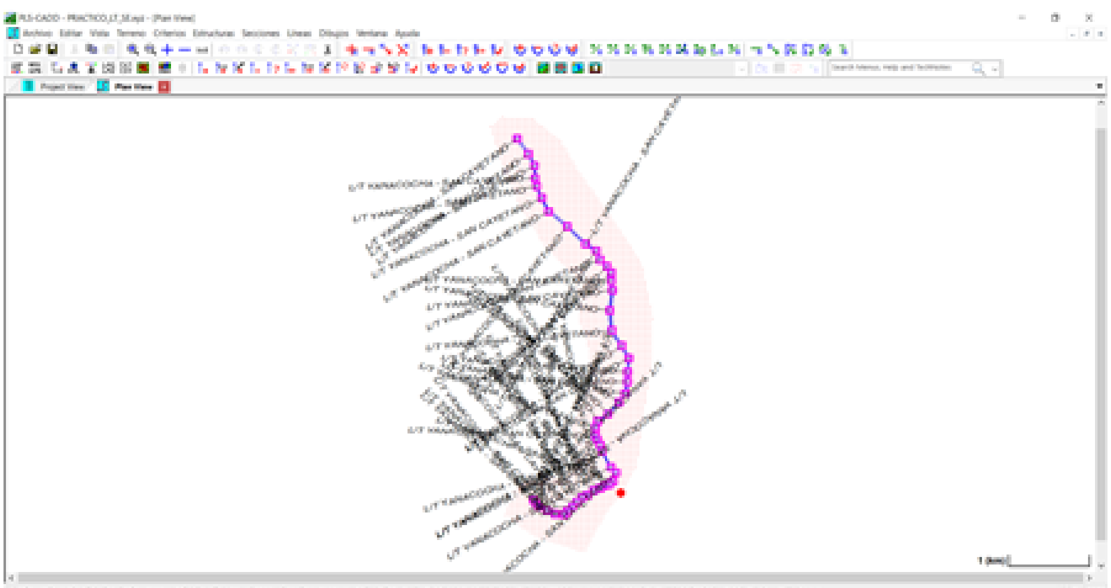


Figure 14: Terrain dataset and preliminary KMZ route displayed together in PLS-CADD.

A new alignment is created using the *Terrain > Alignment* command. The line vertices are digitized along the preliminary route; their positions can be modified later if required. After the unneeded terrain points are removed, the reduced dataset shown in Fig. 15 is obtained.

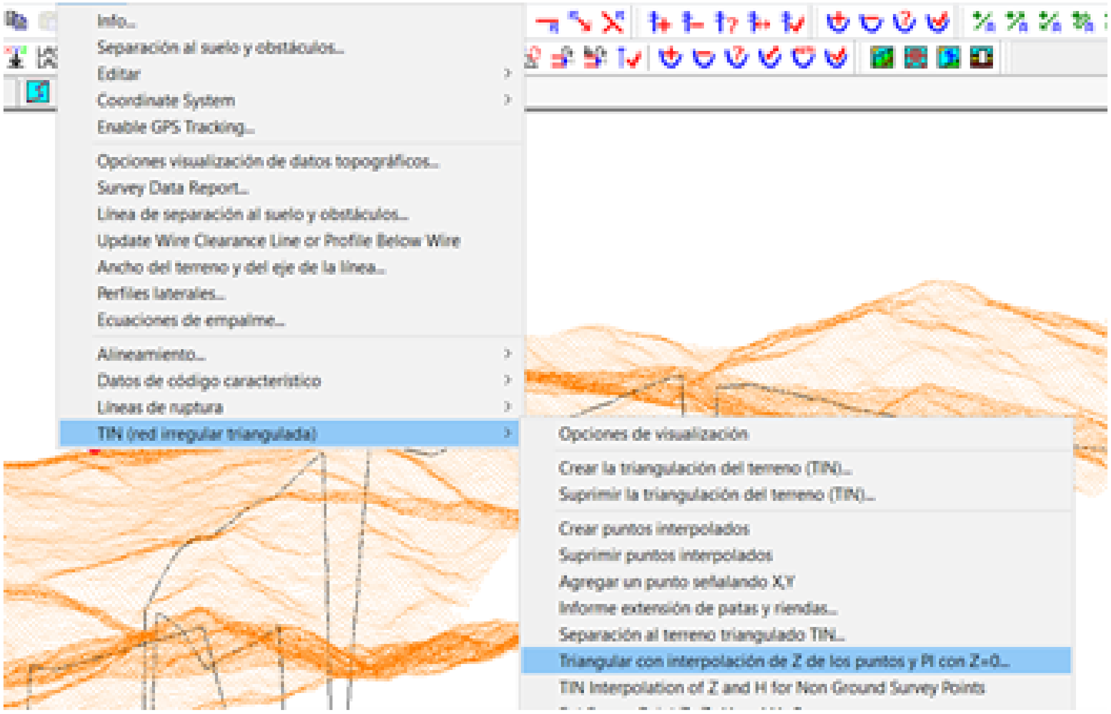


Figure 15: Triangulated terrain and alignment after removal of unnecessary points.

When the model is first displayed in 3-D, elevation errors and misplaced points may be present. Terrain triangulation is therefore performed. In the triangulation dialog, points with $Z = 0$ are excluded, as shown in Fig. 16.

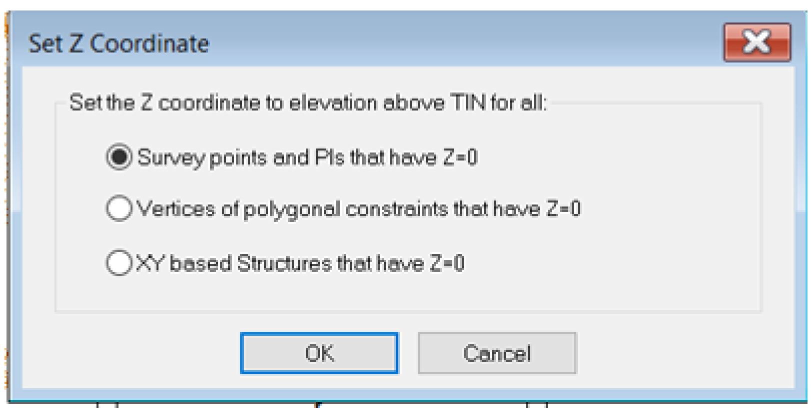


Figure 16: Exclusion of terrain points with zero elevation during triangulation.

Satellite images are then inserted to provide geographic context. In the *Drafting* tool, the *Attachment Manager* is opened. The Web Map Service (WMS) connection is verified and imagery is imported for a 400-m-wide corridor along the alignment, using the settings shown in Fig. 17.

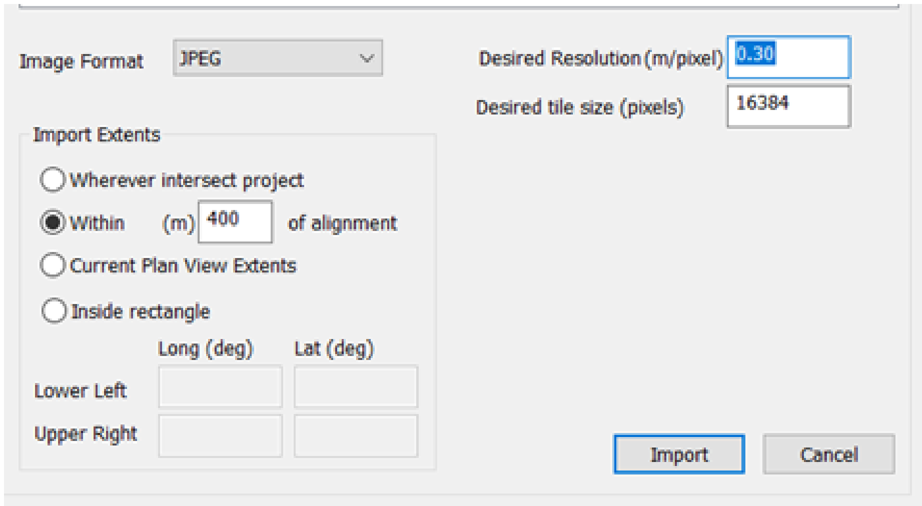


Figure 17: Satellite-imagery import settings for a 400-m alignment corridor.

At this stage, the principal terrain and alignment data have been incorporated into the project. Figures 18–20 show the plan, profile, and 3-D views generated by PLS-CADD.

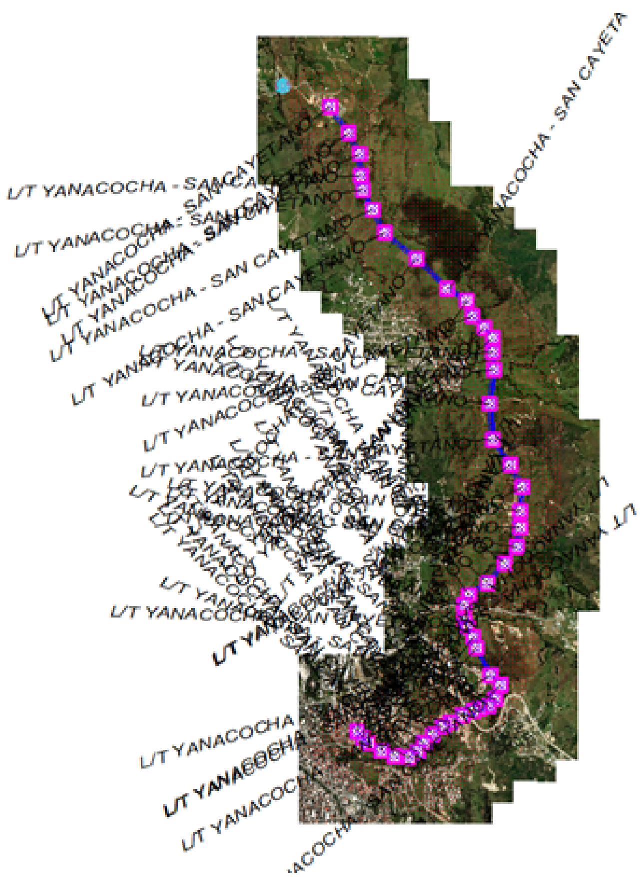


Figure 18: Plan view of the line alignment with satellite imagery.

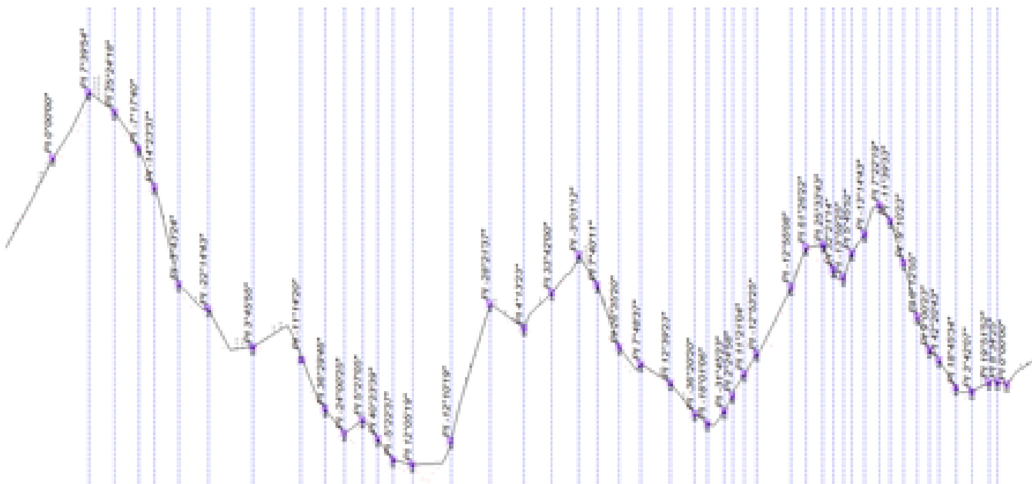


Figure 19: Longitudinal profile view of the preliminary alignment.

Figure 20: Three-dimensional view of the preliminary alignment and terrain model.

2.5 Selection of a 345-kV Transmission Tower

Because 345 kV is not a standardized transmission voltage in Ecuador, an external design reference was consulted. The reference presents the basic geometry of a 345-kV transmission tower, reproduced in Fig. 21 [3]. The original dimensions use U.S. customary units and must consequently be converted to metres before the tower is modeled.

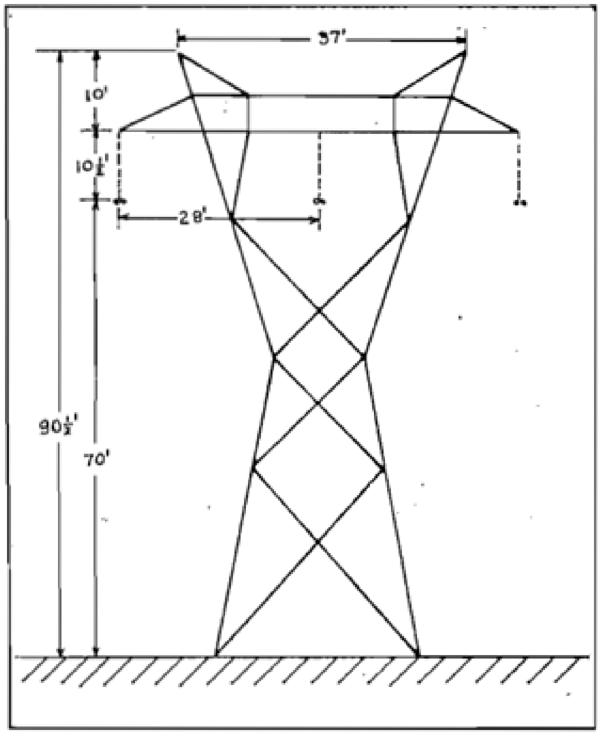
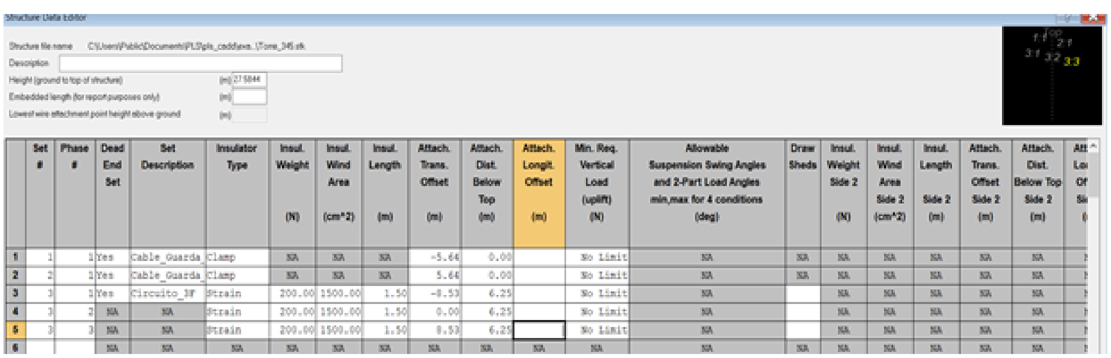


Figure 21: Reference geometry for the 345-kV transmission tower [3].

Before the tower is parameterized, the loading conditions to which the line will be subjected are defined. The built-in criteria file `demo2.cri` is loaded through the *Criteria* tool, and its weather cases are reviewed using the *Weather* command.

The structure is created with the *Structures* tool. The converted dimensions from the reference tower are entered in the structure-definition window shown in Fig. 22; the properties and numbering of each attachment set must be retained correctly.



	Set #	Phase #	Dead End Set	Set Description	Insulator Type	Insul. Weight (N)	Insul. Wind Area (cm^2)	Insul. Length (m)	Attach. Trans. Offset (m)	Attach. Dist. Below Top (m)	Attach. Longit. Offset (m)	Min. Req. Vertical Load (uplift) (N)	Allowable Suspension Swing Angles and 2-Part Load Angles min,max for 4 conditions (deg)	Draw Sheds	Insul. Weight Side 2 (N)	Insul. Wind Area Side 2 (cm^2)	Insul. Length Side 2 (m)	Attach. Trans. Offset Side 2 (m)	Attach. Dist. Below Top Side 2 (m)
1	1	1	Yes	Cable_Guarda_	Clamp	NA	NA	NA	-5.64	0.00		No Limit	NA	NA	NA	NA	NA	NA	NA
2	2	1	Yes	Cable_Guarda_	Clamp	NA	NA	NA	5.64	0.00		No Limit	NA	NA	NA	NA	NA	NA	NA
3	3	1	Yes	Circuito_3F	Strain	200.00	1500.00	1.50	-8.53	6.25		No Limit	NA		NA	NA	NA	NA	NA
4	3	2	NA	NA	Strain	200.00	1500.00	1.50	0.00	6.25		No Limit	NA		NA	NA	NA	NA	NA
5	3	3	NA	NA	Strain	200.00	1500.00	1.50	8.53	6.25		No Limit	NA		NA	NA	NA	NA	NA
6			NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

Figure 22: PLS-CADD structure-definition table for the 345-kV tower.

3 Results

The remaining design activities consist of placing the towers, stringing the conductors, and verifying the resulting line model.

3.1 Initial Trial

An initial trial is performed using a limited number of towers, as shown in Fig. 23. This trial provides a controlled segment in which the stringing configuration can be checked.

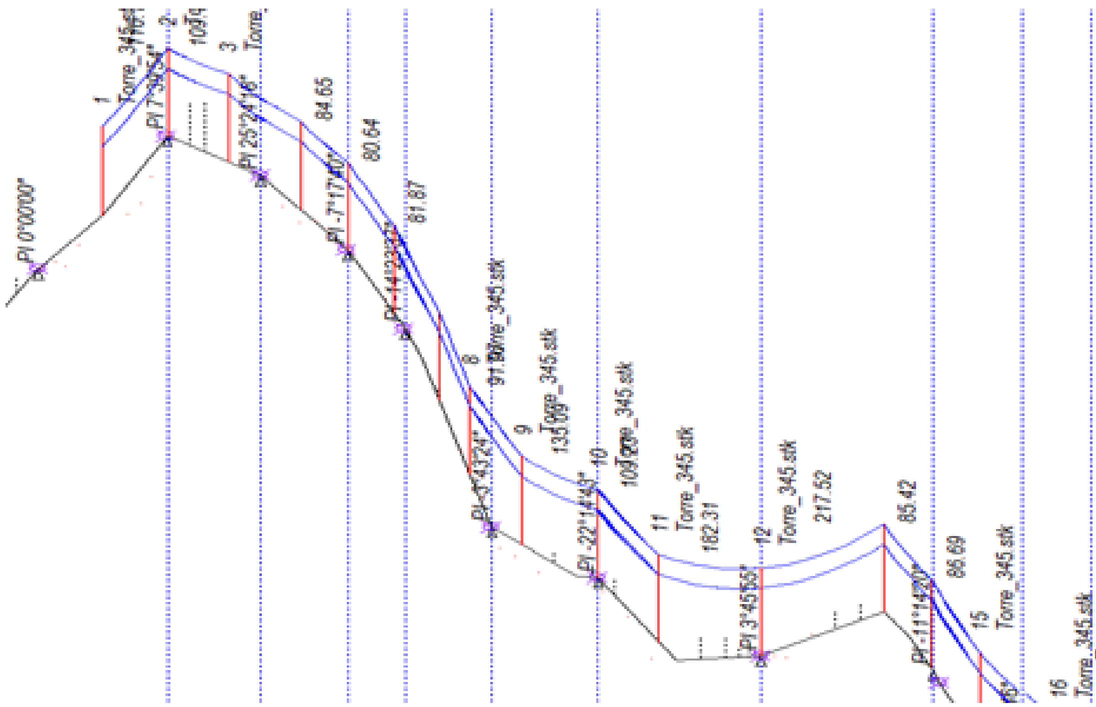


Figure 23: Profile of the initial trial section with a limited number of towers.

The conductors are installed using *Sections > Automatic Sagging*. The corresponding dialog is shown in Fig. 24. For this initial trial, all attachment groups or sets are selected, and the Kiwi conductor available in the default software library is assigned in the *Cable File Name* field.

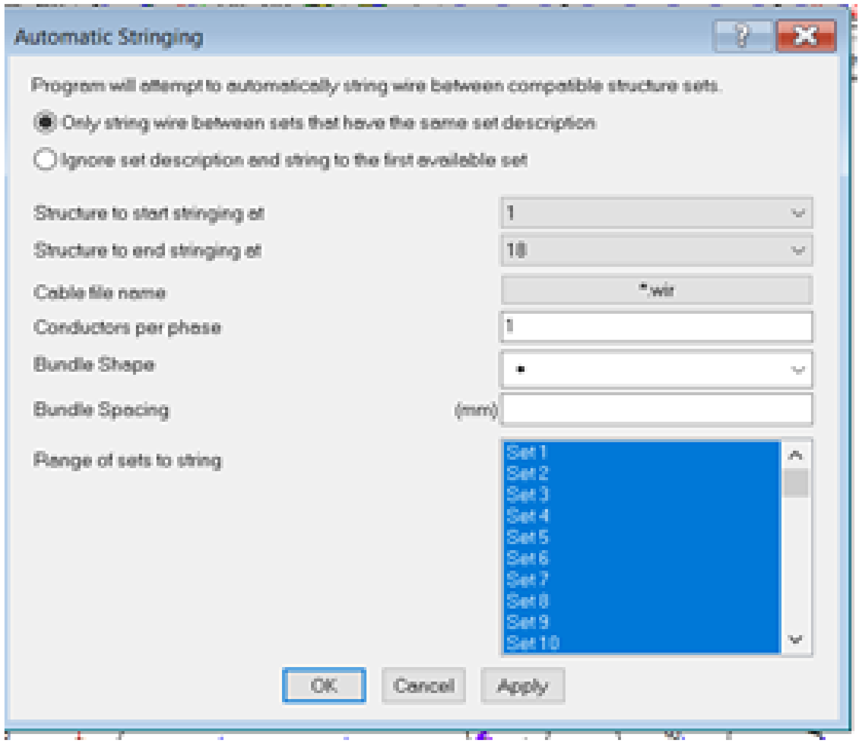


Figure 24: Automatic sagging dialog used for the initial conductor-stringing trial.

The initial result exhibits excessive conductor sag. The section parameters are therefore corrected in the section table shown in Fig. 25. The required fields include the cable name, a nominal voltage of 345 kV, the *NESC Tension Limit (261H1b)* weather case, the *Initial RS* condition, the display color, and, most importantly, the *AutoSag* command.

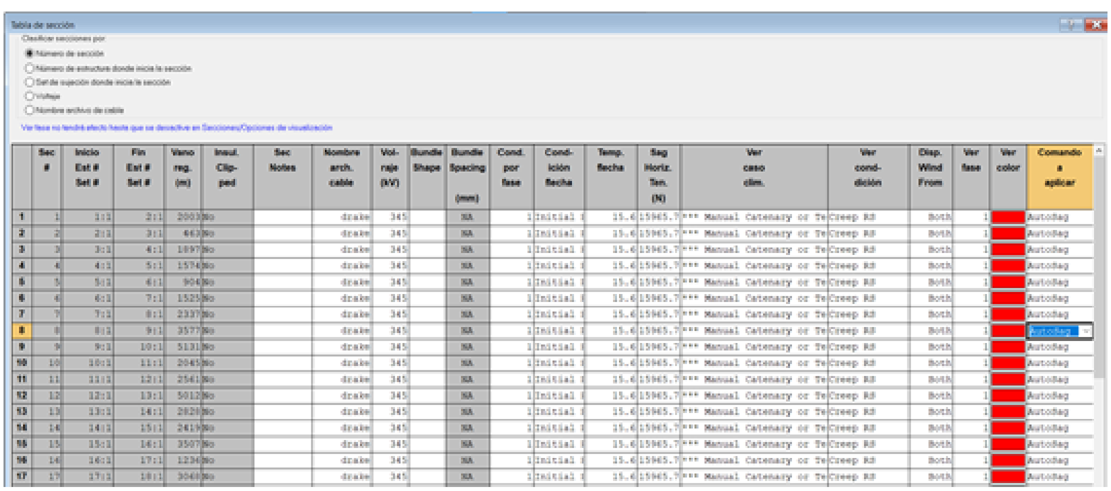


Figure 25: Section table used to assign the conductor and AutoSag parameters.

After these settings are applied, the conductor tension is corrected and the sag is reduced. Figure 26 shows the resulting preliminary model, in which two overhead shield wires, three phase bundles with two subconductors each, and the corresponding insulator strings can be distinguished.

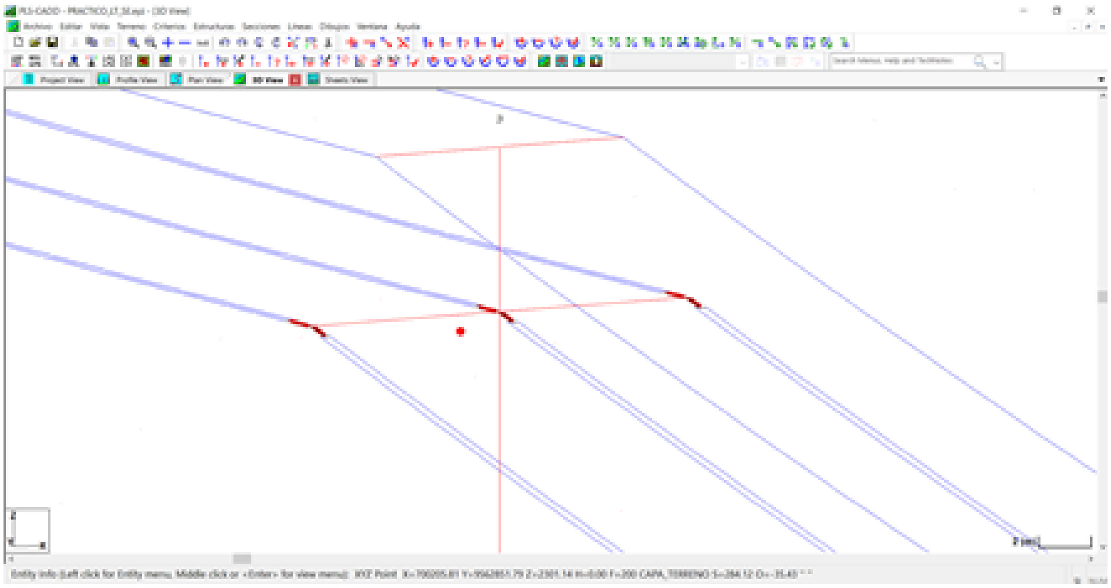


Figure 26: Detail of the corrected initial stringing trial.

3.2 Final Trial

Unlike the initial trial, the final automatic-stringing operation must account for the attachment sets assigned to the shield wires (sets 1 and 2). Two independent stringing operations are required: one for the phase conductors and another for the shield wires, because the latter use a single conductor. Figure 27 shows the relevant PLS-CADD operation.

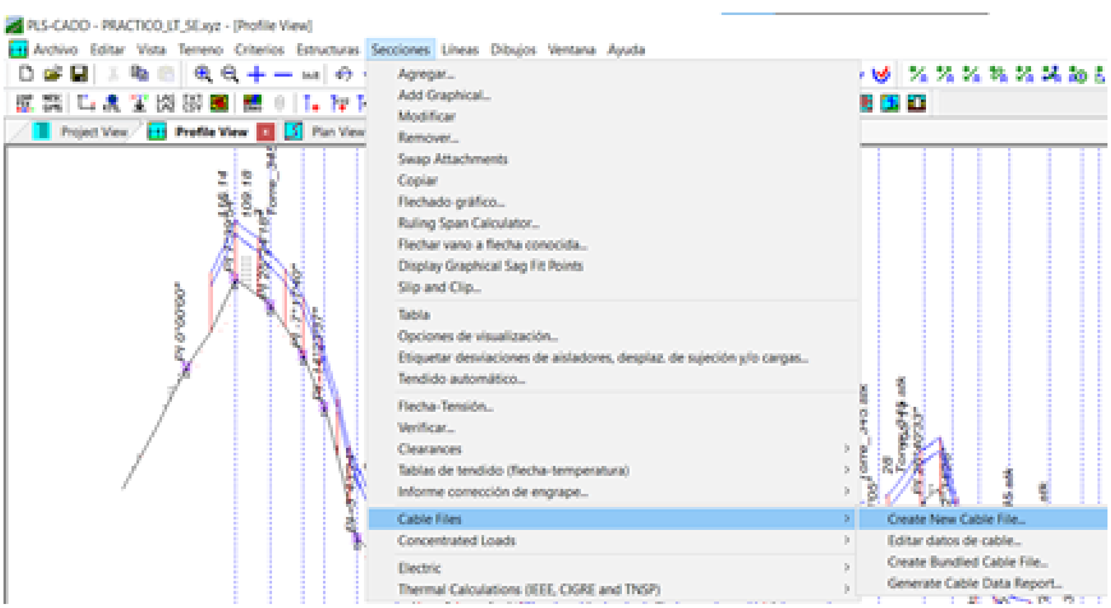


Figure 27: Selection of the line sections for the final stringing operation.

Although any conductor in the software library could be selected, a new cable is created for this project using the cable-definition procedure shown in Fig. 28. The program requires, among other parameters, the conductor cross-sectional area, diameter, unit weight, and rated tensile strength.

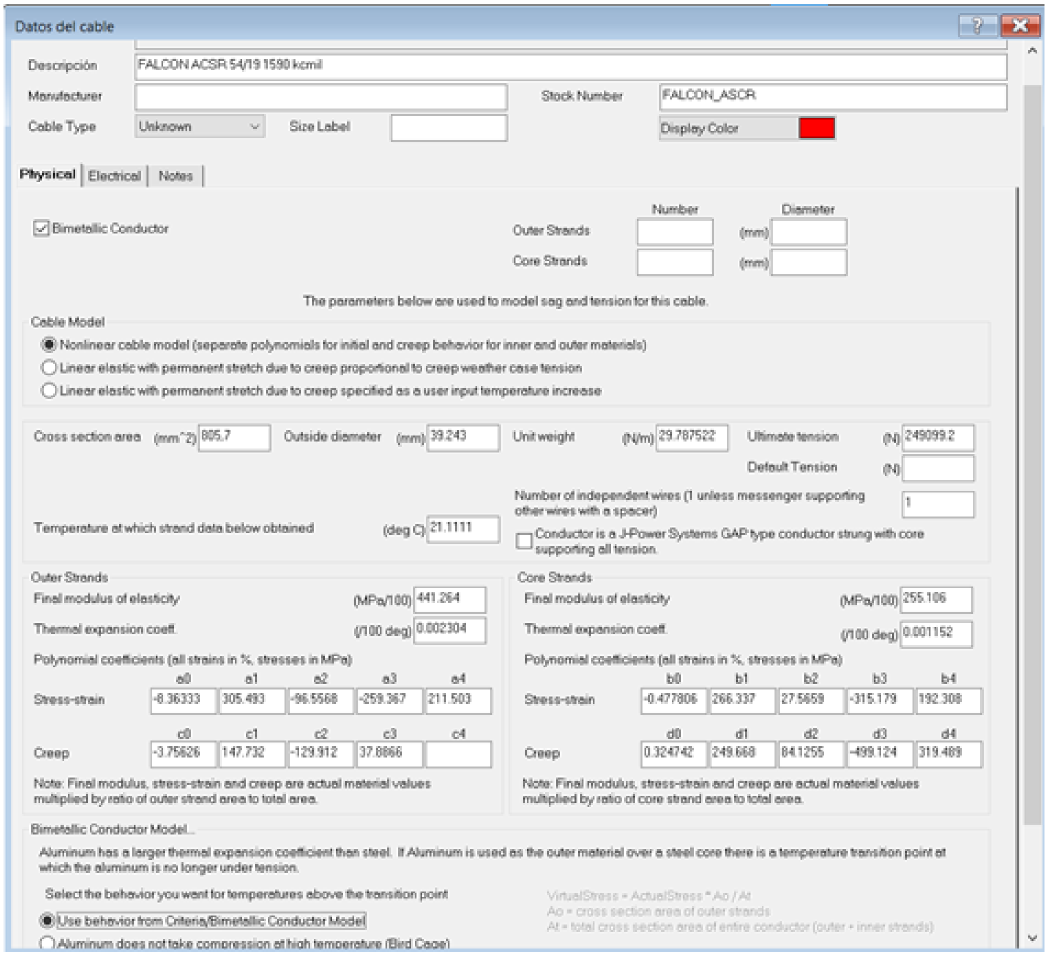


Figure 28: Cable-data dialog used to define a new conductor in PLS-CADD.

The Falcon conductor is selected based on the data reported in [4]. Its principal characteristics are summarized in Table 2.

Table 2: Falcon conductor data used in the model

Parameter	Value
Conductor size (circular mils)	1,590,000
Aluminium strands	54
Steel strands	19
Outside diameter	1.545 in
Rated tensile strength	56,000 lbf
Weight	1,077 lb/mi
Cross-sectional area	805.7 mm^2

The input data are converted to SI units as follows. The rated tensile strength is

$$T = F_{\text{lbf}}(4.4482), \tag{1}$$
$$= 56{,}000(4.4482), \tag{2}$$
$$= 249{,}099.2 \text{ N}. \tag{3}$$

The unit weight is

$$w = w_{\text{lb/mi}}(4.4482)\left(\frac{1}{1609.34}\right), \tag{4}$$
$$= 10{,}777(4.4482)\left(\frac{1}{1609.34}\right), \tag{5}$$
$$= 29.7875 \text{ N/m}. \tag{6}$$

The outside diameter is

$$d = d_{\text{in}}(25.4), \tag{7}$$
$$= 1.545(25.4), \tag{8}$$
$$= 39.243 \text{ mm}. \tag{9}$$

Insufficient information was available for the remaining fields requested by the program. Therefore, the corresponding values of the Drake conductor were adopted as reference parameters. The resulting cable definition is shown in Fig. 28.

Finally, the prescribed conductor-tension criteria are entered through *Criteria* > *Cable Tensions*. Figure 29 presents the weather cases, cable conditions, and percentage limits applied in the final design.

	Weather Case	Cable Condition	% de Tensión Última
1	NESC Medium District Loading (250B)	Initial RS	40.000
2	NESC Extreme Wind (250C)	Initial RS	60.000
3	NESC Concurrent Ice and Wind (250D)	Initial RS	60.000
4	NESC Tension Limit (261H1b)	Initial RS	35.000
5	NESC Tension Limit (261H1b)	Creep RS	30.000

Figure 29: Conductor tension criteria adopted for the final line model.

3.3 Final Transmission-Line Model

The completed transmission-line model is presented in profile, plan, 3-D, and plan-and-profile sheet views. The longitudinal profile in Fig. 30 shows the support locations, conductor catenaries, terrain profile, and clearance relationships along the route.

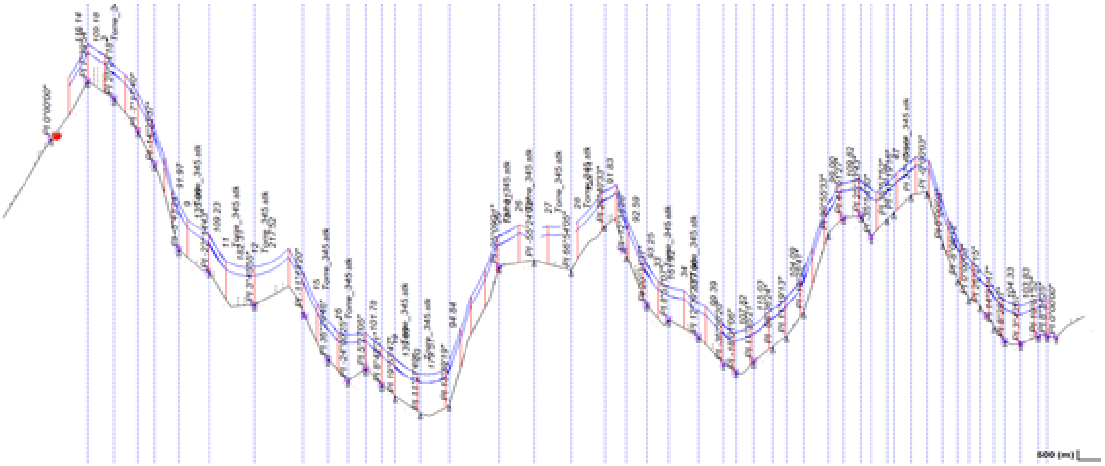


Figure 30: Longitudinal profile of the completed transmission line.

The plan view with satellite imagery is shown in Fig. 31, while Figs. 32 and 33 show the line geometry without the imagery layer at different scales.

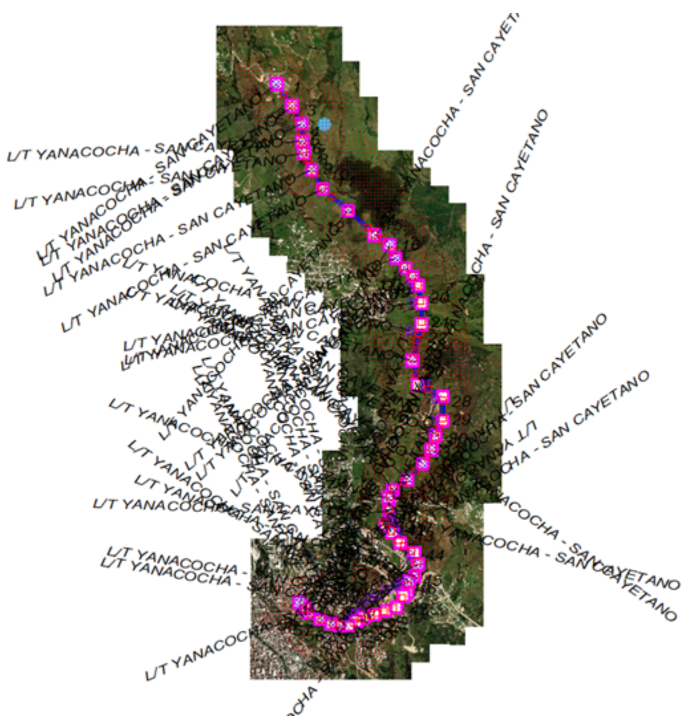


Figure 31: Plan view of the completed transmission line with satellite imagery.

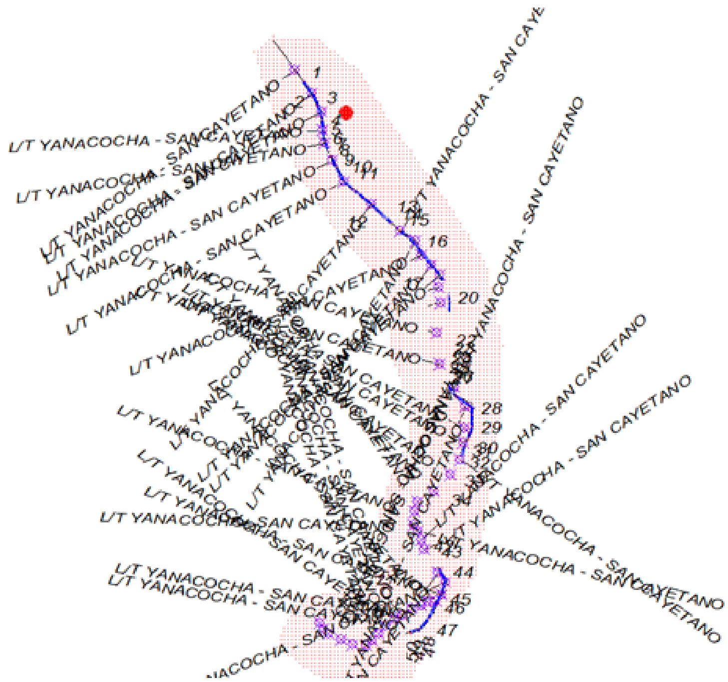


Figure 32: Overall plan view of the completed transmission-line route.

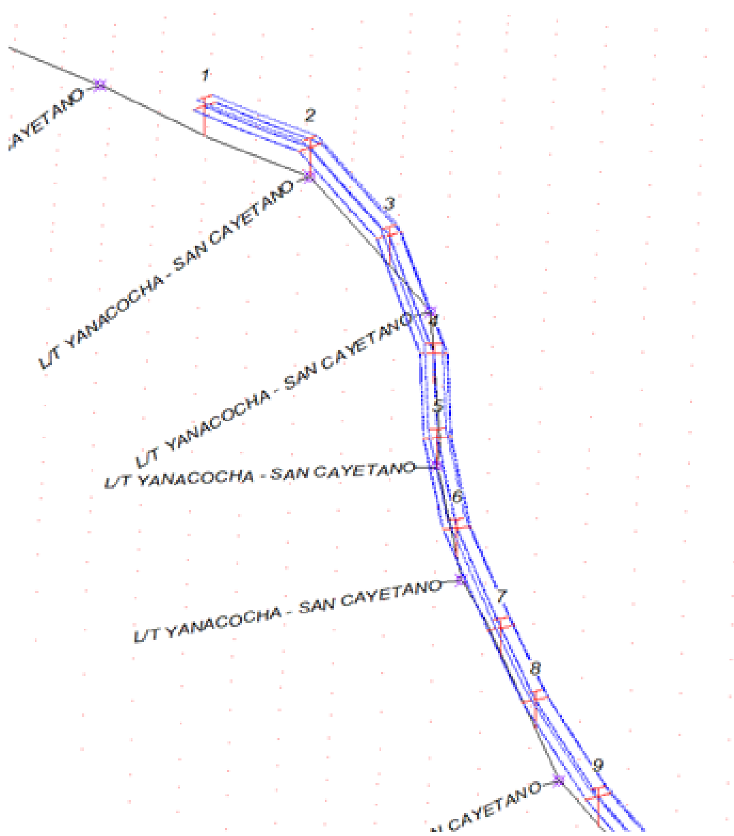


Figure 33: Detailed plan view of the southern section of the route.

Figure 34 presents a detailed 3-D view of a representative portion of the modeled line.

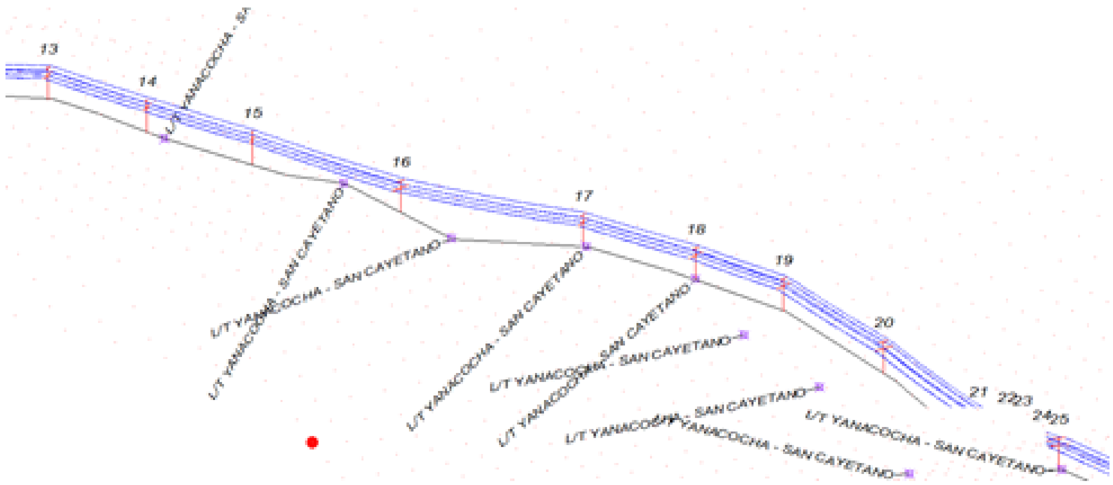


Figure 34: Three-dimensional detail of the completed transmission-line model.

Figures 35–39 show the final plan-and-profile sheets exported from PLS-CADD. Each sheet combines the georeferenced plan strip with the corresponding terrain, structure, conductor, and stationing information.

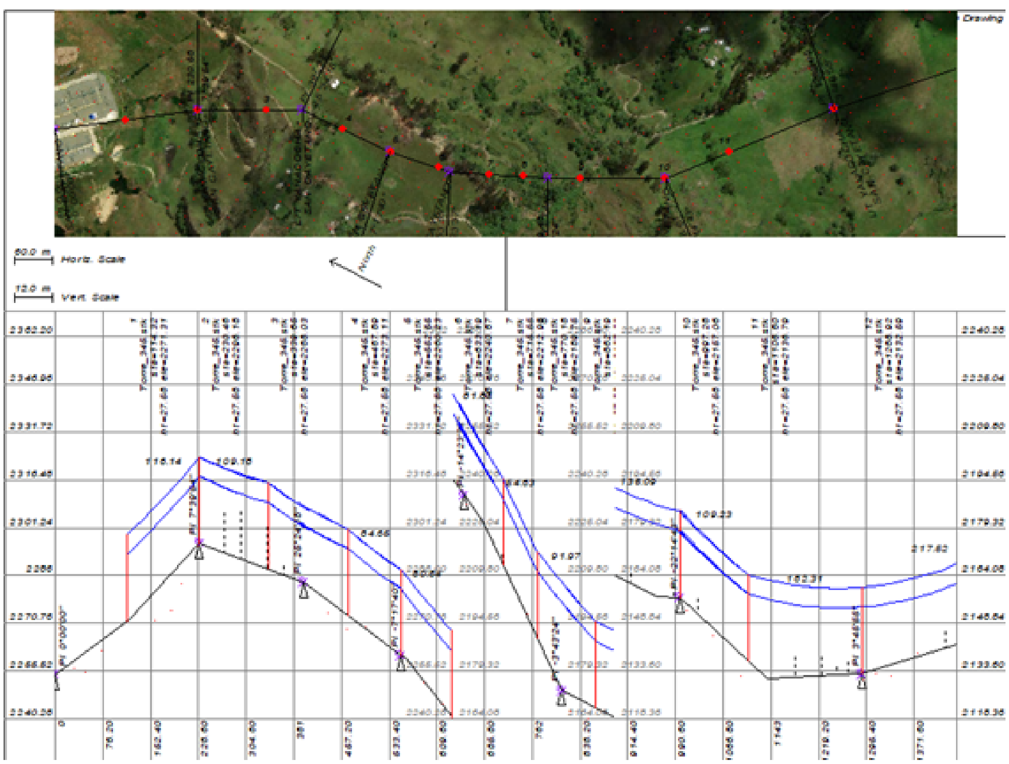


Figure 35: Plan-and-profile sheet 1 of the final transmission-line design.

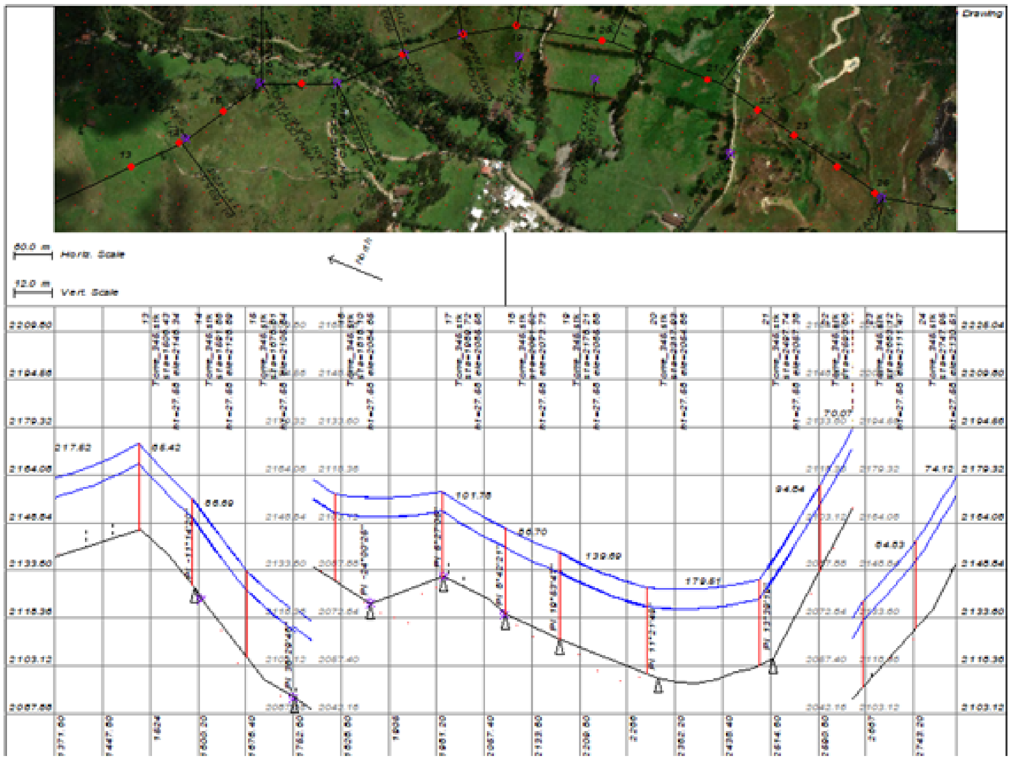


Figure 36: Plan-and-profile sheet 2 of the final transmission-line design.

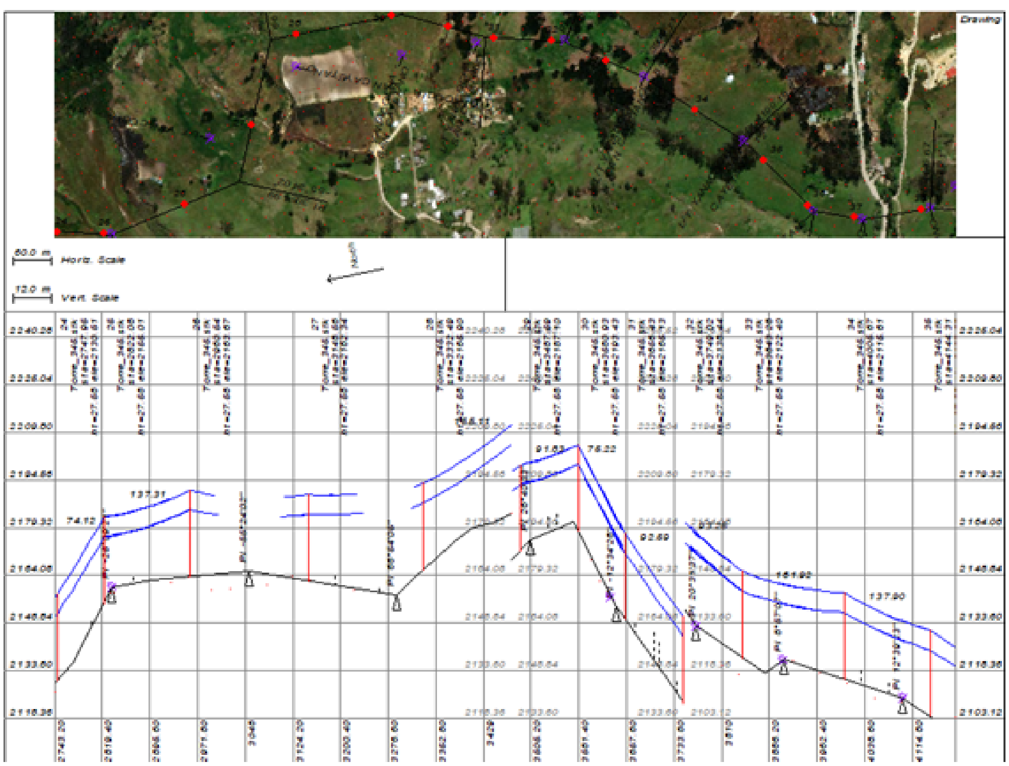


Figure 37: Plan-and-profile sheet 3 of the final transmission-line design.

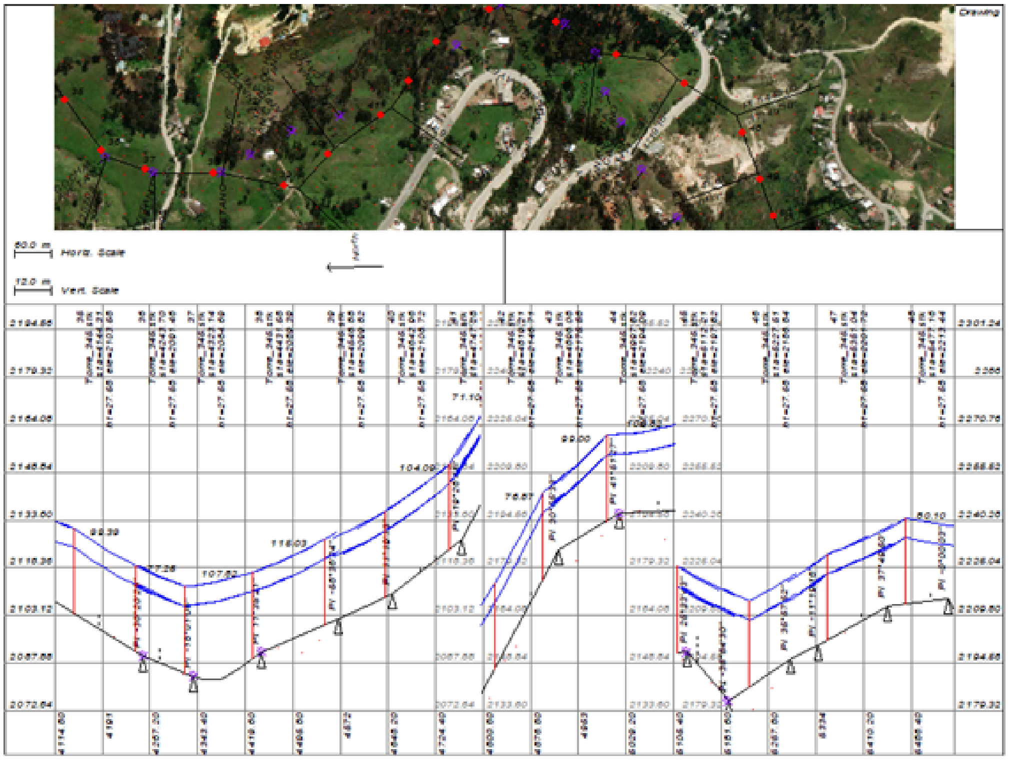


Figure 38: Plan-and-profile sheet 4 of the final transmission-line design.

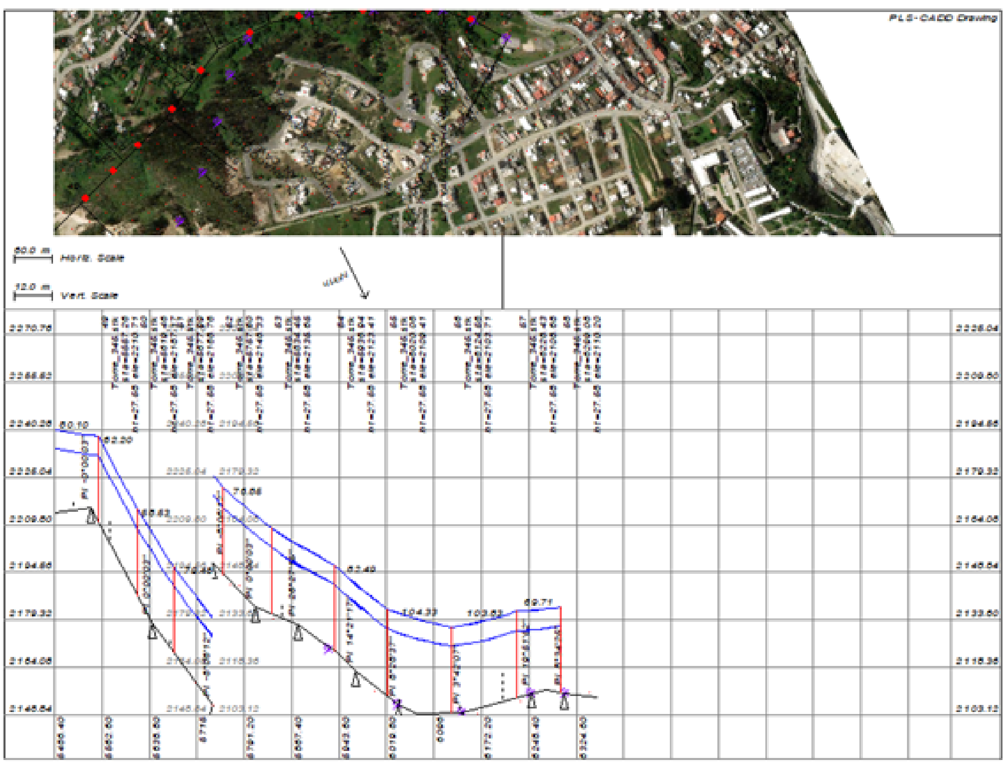


Figure 39: Plan-and-profile sheet 5 of the final transmission-line design.

3.4 Cable Utilization

The detailed cable-utilization results are provided in the accompanying spreadsheet entitled `LTSE_U02_PRTC1_USO_CABLES_GRANDA_LEONARDO`. A separate file is used because PLS-CADD generates a large volume of numerical output for the modeled sections.

References

[1] NASA Earthdata, "Earthdata Search," 2025. [Online]. Available: `https://search.earthdata.nasa.gov/`. [Accessed: Jul. 9, 2025].

[2] Google, "Google Earth." [Online]. Available: `https://www.google.com/earth/`. [Accessed: Jul. 9, 2025].

[3] V. H. O. Luna, *Selección y coordinación de aislamiento para el sistema de transmisión Paute–Guayaquil*. Quito, Ecuador: Escuela Politécnica Nacional, 1972.

[4] P. M. Anderson, *Analysis of Faulted Power Systems*. New York, NY, USA: Wiley-IEEE Press, 1995.